NewsNuggets: Automated News Article Summarization and Categorization

Suryansh Srivastava
Department of Data Science and Artificial Intelligence
Asian Institute of Technology
Pathum Thani, Thailand
st124997@ait.asia

Phue Pwint Thwe
Department of Data Science and Artificial Intelligence
Asian Institute of Technology
Pathum Thani, Thailand
st124784@ait.asia

Abstract

This progress report outlines the development of NewsNuggets, an automated near real-time news summarization and categorization system that utilizes OpenAI's large language models via API. The pipeline is designed to collect news from APIs or web sources, preprocess it, then summarize and categorize the content using OpenAI. The summarized results are stored and served through an Amazon API Gateway to a mobile application built with React Native (TypeScript, EXPO). The system is implemented using a fully serverless architecture powered by AWS Lambda, Kinesis, DynamoDB, and integrated observability through CloudWatch. This design ensures near real-time performance, ease of deployment, and horizontal scalability.

1 Introduction

1.1 Background of the study

With the rapid growth of digital news, users are often overwhelmed by the sheer volume of content. Manually filtering, reading, and categorizing news is inefficient, leading to information overload. Traditional summarization tools rely on extractive methods, which fail to provide cohesive and readable summaries. Similarly, keyword-based classification systems struggle with ambiguous, multi-topic articles. This project aims to create an AI-powered solution that automatically summarizes news articles and categorizes them into predefined topics using advanced NLP techniques. Our approach ensures high accuracy, real-time scalability, and improved user accessibility by integrating cloud-based machine learning into the pipeline.

1.2 Problem Statement

The existing challenges in news summarization and categorization include:

1. **Information Overload:** Users struggle to process large volumes of daily news articles efficiently.

2. **Extractive Summarization Limitations:** Conventional extractive techniques lack readability and fail to paraphrase content.

3. **Inaccurate Categorization:** Rule-based and keyword-based classifiers misclassify news due to semantic ambiguity.

4. **Lack of Automation:** Many news summarization solutions require manual curation or human intervention.

5. **Scalability Issues:** Existing models do not efficiently scale for real-time summarization and categorization across multiple sources.

To address these issues, our system provides automated, abstractive summarization and AI-driven categorization with real-time processing capabilities.

2 Proposed Solution

To address the limitations of traditional news summarization and classification systems, the NewsNuggets platform has been designed with a highly modular, serverless architecture, ensuring real-time processing, operational scalability, and low infrastructure overhead. The system leverages OpenAI's GPT-powered APIs to provide abstractive summarization and topic classification, replacing rule-based, extractive models with semantically rich, coherent summaries.

2.1 Architecture Flow

The architecture is designed for full automation, real-time processing, and seamless frontend integration. The system workflow is as follows in figure 1:

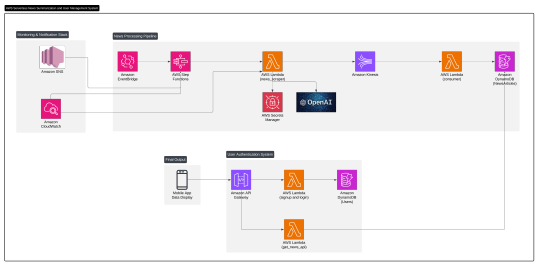

Figure 1: System Architecture for NewsNuggets: End-to-End News Summarization and Categorization

1. User Registration and Authentication: Users interact with the platform by signing up or logging in via a REST API exposed through Amazon API Gateway. These requests are routed to dedicated Lambda functions (signup, login) which interact with a DynamoDB Users table to validate and store user credentials.

2. News Acquisition (Scraping): A scheduled Step Function triggers the news_scraper Lambda function every hour using an Amazon EventBridge rule. This function fetches the latest news articles from a defined source, performs preliminary text cleaning, and pushes the data into a real-time Kinesis data stream.

3. Streaming and Real-time Processing: The Kinesis stream, news-stream, acts as the ingestion layer for scraped content. It forwards each incoming news article to a downstream consumer Lambda function (consumer), using event source mapping.

4. Summarization and Categorization: The consumer Lambda function processes each Kinesis record, invokes OpenAI's API to generate abstractive summaries and predict article categories, and stores the processed records in the NewsArticles table in DynamoDB.

5. Frontend Data Access: The system exposes a RESTful GET /getnews endpoint via API Gateway. This endpoint invokes the get_news_api Lambda, which queries DynamoDB for the latest articles and returns them to the React Native frontend in a clean, filterable format.

6. Monitoring and Alerting: Amazon CloudWatch logs and monitors all Lambda functions and Step Function executions. An alerting system using Amazon SNS notifies developers via email if the Step Function fails even after retries, ensuring rapid incident response and operational continuity.

This architectural design supports high availability, auto-scaling, and asynchronous processing while ensuring secure and efficient delivery of summarized news content to end users.

3 Related Works

A number of systems have been developed for news summarization and categorization, ranging from industry tools to academic research.

Traditional platforms such as Google News, Flipboard, SummarizeBot, and IBM Watson Discovery primarily rely on extractive summarization methods, which select and present key sentences from the original content. These approaches often use rule-based or statistical keyword extraction techniques that may result in fragmented and less readable summaries.

With advancements in Natural Language Processing, recent academic research has focused on abstractive summarization using Transformer-based models such as BART (Lewis et al., 2020) and PEGASUS (Zhang et al., 2020). These models are capable of generating fluent, paraphrased summaries rather than merely extracting existing sentences. PEGASUS, for instance, is pre-trained using a gap-sentence generation objective tailored for news content.

Classification models such as DistilBERT (Sanh et al., 2019) and RoBERTa (Liu et al., 2019) have been widely used for topic categorization and semantic understanding tasks. However, they typically require infrastructure for training, fine-tuning, and maintenance.

In contrast,our proposed system NewsNuggets utilizes the OpenAI API (OpenAI, 2023) to perform both summarization and categorization in real time. This approach removes the overhead of hosting and managing models, while leveraging the power of state-of-the-art large language models like GPT-4. The integration of OpenAI's capabilities into a serverless, cloud-native architecture ensures scalability, low latency, and simplified operations, making it a more efficient alternative to traditional model deployment strategies.

4 Methodology

This section outlines the complete methodology used to design and implement the NewsNuggets

system, including the end-to-end data pipeline, preprocessing stages, service integrations, and evaluation strategy. The approach prioritizes real-time performance, modular service structure, security, and ease of scalability using AWS cloud-native services and OpenAI's inference API.

4.1 Data Pipeline and Preprocessing

The system's data pipeline begins with the automated scraping of news articles from public APIs or web sources. A serverless Lambda function handles data ingestion and preprocessing tasks such as HTML tag removal, whitespace trimming, and encoding normalization. Cleaned articles are streamed in real time using Amazon Kinesis to a downstream consumer Lambda function, which performs semantic summarization and topic categorization using the OpenAI API.

Summarized content is stored in DynamoDB, from which the frontend mobile app (developed with React Native and TypeScript using EXPO) retrieves and displays the data via API calls. All access credentials, including OpenAI API keys, are securely stored using AWS Secrets Manager. The pipeline is monitored via Amazon CloudWatch for logs and metrics, while Amazon SNS provides failure alerts for critical operations like Step Functions.

The pipeline comprises the following stages:

1. Scraping and cleaning using AWS Lambda.
2. Real-time streaming via AWS Kinesis.
3. Summarization and categorization using OpenAI API.
4. Storage in Amazon DynamoDB.
5. Data serving through Amazon API Gateway.
6. Mobile display using a React Native (TypeScript, EXPO) app.

4.2 Services and Their Purpose

The implementation of NewsNuggets relies on a suite of AWS services, each playing a specific role in facilitating automation, real-time data handling, system scalability, and secure information management. The components described below were carefully selected to support modular development and cloud-native deployment practices.

1. AWS Lambda serves as the main computational engine for the platform. It includes several stateless functions:
   - The `news_scraper` Lambda is responsible for collecting the latest news content from online sources. Its execution is automated and scheduled on an hourly basis using AWS Step Functions in combination with EventBridge.
   - The `consumer` Lambda processes incoming article records streamed through Kinesis. It calls the OpenAI API to perform summarization and categorization, then stores the results in DynamoDB.
   - The `get_news_api` Lambda handles frontend requests made to the `GET /getnews` endpoint. It retrieves the most recent summarized content from the `NewsArticles` table.
   - Two additional Lambda functions, `signup` and `login`, manage user registration and authentication by interacting with the `Users` table in DynamoDB.
2. API Gateway (v2 - HTTP API) provides the external RESTful interface for the mobile frontend. It routes incoming requests to the appropriate Lambda functions. Three primary endpoints are defined:
   - `GET /getnews`, which connects to the `get_news_api` Lambda
   - `POST /signup`, which connects to the `signup` Lambda
   - `POST /login`, which connects to the `login` Lambda

   The gateway also has Cross-Origin Resource Sharing (CORS) enabled to allow interaction with the mobile frontend across domains.
3. Amazon DynamoDB is used as the NoSQL storage system. It contains two primary tables:
   - The `Users` table maintains user credentials and uses a unique `id` as the partition key.
   - The `NewsArticles` table stores summarized news content. It is indexed by `article_id` (partition key) and `publishedAt` (sort key) to allow time-based queries.
4. Amazon Kinesis is used to handle real-time ingestion of scraped news articles. The stream,

named `news-stream`, transfers records from the `news_scraper` Lambda to the `consumer` Lambda using an event source mapping. This setup ensures asynchronous and scalable processing of data.

5. AWS Step Functions orchestrate the execution of the `news_scraper` Lambda. A simple one-state state machine is employed, which includes retry logic configured to attempt up to two re-executions in the event of a failure, with a five-minute delay between attempts.

6. EventBridge is responsible for triggering the Step Function at regular intervals. Currently, the system is configured to run once every hour (`rate(1 hour)`), although this schedule can be easily adjusted depending on evolving system requirements.

7. Amazon CloudWatch provides comprehensive monitoring and logging support. It captures execution logs for all Lambda functions and Step Function invocations and monitors key metrics such as the `ExecutionsFailed` count to ensure operational visibility and resilience.

8. Amazon SNS (Simple Notification Service) is used to send real-time alerts. A topic named `step-function-failure-alerts` is configured to notify the system administrator via email (`suryanshsrivastava22@gmail.com`) when the Step Function fails, even after retries.

9. AWS Secrets Manager ensures the secure handling of credentials. It stores the OpenAI API key under the name `apikeys`, allowing it to be safely accessed by authorized Lambda functions without exposing secrets in the codebase or environment variables.

5 Experimental Results

As part of our initial implementation, we have successfully completed and tested several components of the NewsNuggets pipeline.

A Lambda function (news scraper) was created and deployed using ARM64 architecture to scrape the latest articles from the Bangkok Post website. The scraped content is cleaned and sent to a Kinesis stream named news-stream. The figure 2 below illustrates the deployed function and its scraping logic

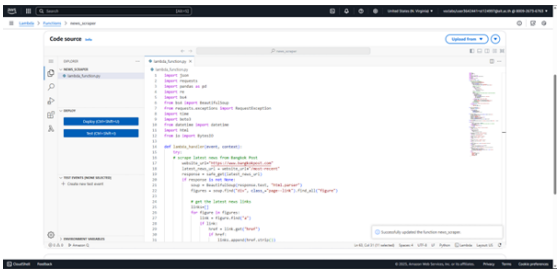

Figure 2: Deployed Lambda function news scraper scraping from Bangkok Post and sending to Kinesis

The news-scraper Lambda is the entry point of the pipeline. It scrapes news from the latest BKK Post page, cleans the data, and streams it to Kinesis. This triggers a consumer Lambda function that retrieves the data from Kinesis and stores it in DynamoDB.

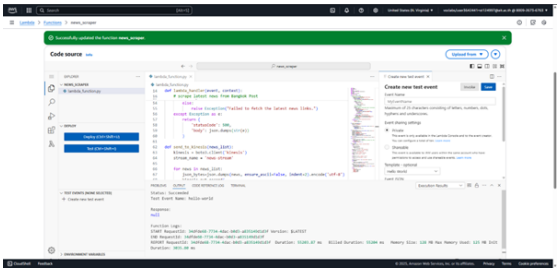

Figure 3: Scraper Lambda invoked and data successfully fed to Kinesis

A Kinesis data stream named news-stream was created to receive and stream the incoming article data. The below figure 4 confirms successful creation and active status of the stream.

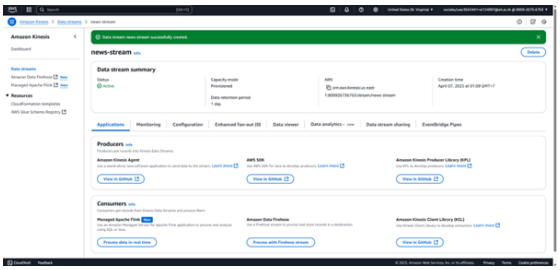

Figure 4: Amazon Kinesis stream news-stream for real-time ingestion

Additional view from the Kinesis stream confirms that scraped articles are present in the stream buffer.

A DynamoDB table named NewsArticles was created with article-id as the partition key and publishedAt as the sort key, optimized for querying time-based article entries. The figure 6 below shows the DynamoDB configuration.

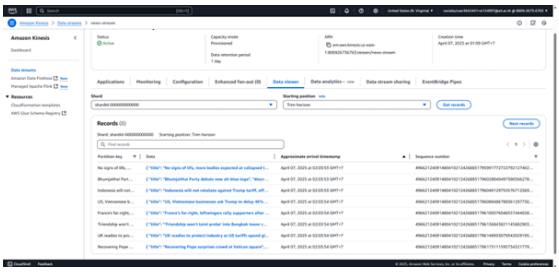

Figure 5: Kinesis Data Viewer displaying streamed news records from Lambda scraper

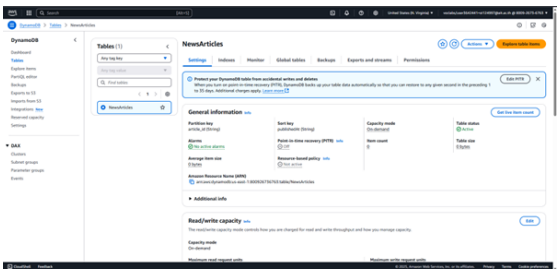

Figure 6: DynamoDB table NewsArticles with partition key

The consumer Lambda function is automatically triggered to read the stream and insert records into DynamoDB. The table shows successful storage of summarized entries.

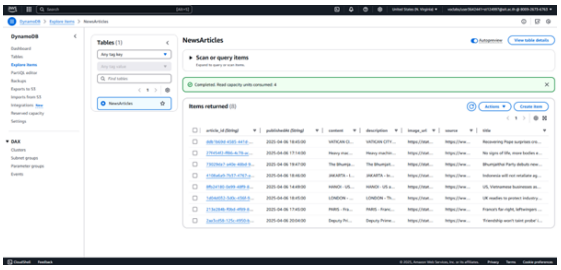

Figure 7: DynamoDB table NewsArticles showing successfully stored entries

The following screenshot presents one complete article record stored in DynamoDB with all required fields.

A consumer Lambda function has been partially implemented. It reads data from Kinesis and stores it in DynamoDB. The function is also intended to call OpenAI's API for summarization and categorization. Although this step has already been tested separately, it has not yet been integrated into the consumer Lambda function. The function securely fetches the OpenAI API key from AWS Secrets Manager. The below figure 9 displays the consumer function setup.

The OpenAI API was tested independently using

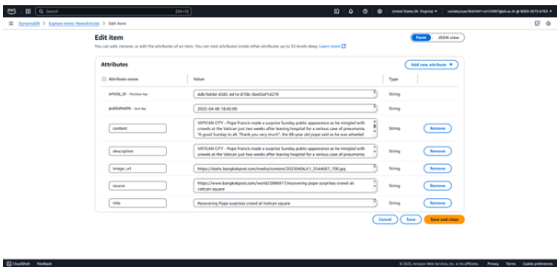

Figure 8: Detailed view of the first article stored in DynamoDB

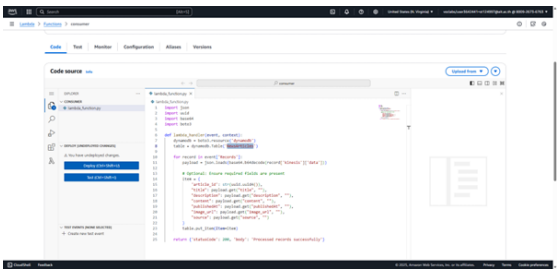

Figure 9: Consumer Lambda function prepared to process from Kinesis to DynamoDB and connect to OpenAI API

sample article prompts. The API returned both the summary and the predicted category, confirming its reliability. Below are screenshots from Postman showing the test prompt and the successful response returned by OpenAI.

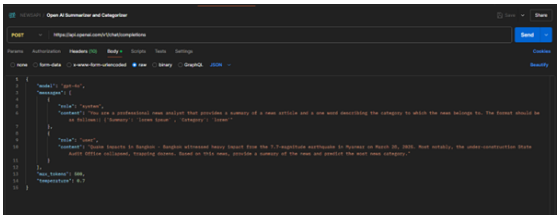

Figure 10: OpenAI summarization and categorization API test with input prompt

Preliminary logs and screenshots show successful data flowing from the scraper to Kinesis, then from Kinesis to DynamoDB, validating end-to-end communication for unprocessed articles.

The next steps involve full integration of OpenAI's API into the consumer Lambda function and expanding support for multilingual and geolocation-based filtering. Overall, the groundwork for a scalable, serverless summarization pipeline has been laid.

Figure 11: OpenAI API response: generated summary and predicted category

6 Architecture: AWS Well-Architected Framework

The revised architecture of NewsNuggets is carefully aligned with the principles of the AWS Well-Architected Framework. Operational excellence is achieved through the use of Infrastructure as Code (IaC) via Terraform, along with modular Lambda functions that allow for clean, repeatable deployments. Security is enforced through AWS Identity and Access Management (IAM) roles and Secrets Manager, which help ensure that access permissions are tightly controlled and sensitive credentials are securely stored. The system is designed for reliability, leveraging AWS Kinesis for real-time streaming and incorporating retry mechanisms through AWS Lambda to handle transient failures and maintain consistent data processing. Performance efficiency is optimized by using lightweight Lambda functions that asynchronously invoke the OpenAI API, reducing latency and ensuring smooth scalability. Finally, cost optimization is addressed by relying entirely on serverless services, which automatically scale with demand and eliminate the need for overprovisioned infrastructure. This architecture enables a highly responsive, secure, and cost-effective solution for real-time news summarization and categorization.

7 Scalability Considerations

The NewsNuggets system is designed to efficiently scale across varying data loads and user demands through the use of fully managed, serverless AWS services. API calls to fetch summaries are routed through AWS Lambda maintaining an average latency of under 500 milliseconds per request. This ensures real-time responsiveness even during traffic spikes.

The system's streaming layer, powered by AWS Kinesis, is capable of handling between 50,000 to 100,000 news articles per day. This allows ingestion of large-scale, continuous news data feeds with minimal performance degradation.

Additional scalability is achieved through efficient data storage design. DynamoDB is used to store structured summaries, supporting high-throughput read/write access. The serverless architecture ensures that resources are only consumed as needed, allowing the system to scale automatically without manual intervention.

The scalability features can be summarized as follow:

1. Serverless compute (Lambda) enables automatic scaling.

2. DynamoDB and Kinesis scale with incoming data and read/write volume.

3. API latency remains low through asynchronous processing.

4. OpenAI API offloads summarization, eliminating local compute demands.

5. The system can handle hundreds of articles daily with minimal infrastructure overhead.

8 Future Work

Following the successful implementation of the core NewsNuggets pipeline, the next phase of development will focus on enhancing system usability, observability, personalization, and deployment automation. These improvements will build upon the existing modular and serverless architecture that incorporates AWS Lambda, API Gateway, Kinesis, DynamoDB, Step Functions, EventBridge, CloudWatch, SNS, and Secrets Manager.

To provide more relevant content, the system will support geolocation-based filtering, enabling users to receive news summaries tied to their location. This will require extending the consumer Lambda to process and store geolocation metadata within the NewsArticles DynamoDB table. Additionally, multilingual support will be added by updating the news_scraper Lambda to fetch content from non-English sources and testing OpenAI's capability to handle summarization in languages such as Thai.

A significant long-term goal is to reduce dependency on third-party APIs by training a custom summarization and classification model. This model will initially be developed and uploaded to

an S3 bucket along with its weight files. Over time, as the system accumulates labeled data in the DynamoDB table—consisting of user-rated summaries and categorized articles—a fine-tuning pipeline can be implemented. The model will be retrained periodically on this dataset to improve its performance and alignment with user expectations.

This direction will allow the project to gradually transition from reliance on OpenAI to a fully self-hosted solution. Although the system will continue using OpenAI in the short term, custom model development will enable long-term autonomy and cost control.

In essence, while NewsNuggets serves users with real-time AI-generated summaries, it simultaneously builds a growing dataset that will support future model training. The project's evolution thus involves both improving the current pipeline and laying the groundwork for an intelligent, adaptable, and independent summarization engine.

These planned improvements aim to evolve NewsNuggets into a fully automated, user-adaptive news summarization platform capable of serving a global audience with personalized content in multiple languages.

9 Conclusion

NewsNuggets provides a comprehensive solution for real-time news summarization and categorization using OpenAI's language models integrated into a fully serverless architecture on AWS. The platform combines core services such as Lambda, Kinesis, DynamoDB, API Gateway, Step Functions, EventBridge, CloudWatch, SNS, and Secrets Manager to deliver a scalable, efficient, and secure system.

The architecture enables scheduled scraping, real-time data streaming, secure user authentication, and low-latency API delivery to a React Native mobile application. By relying on the OpenAI API for inference, the system avoids the complexity of model training and infrastructure management, while Secrets Manager ensures secure handling of credentials.

Step Functions and EventBridge automate the regular execution of the scraping workflow with retry logic, and CloudWatch provides centralized monitoring and performance tracking. Alerts generated via SNS enhance system reliability by notifying administrators of pipeline failures. The system adheres to the AWS Well-Architected Framework, meeting key principles of operational excellence, performance efficiency, security, reliability, and cost optimization.

The current implementation demonstrates successful scraping, streaming, and storage of news articles using AWS services. Although OpenAI API integration is still underway, preliminary tests confirm its capability for high-quality summarization and categorization. Future developments will extend the system with location-aware content, multilingual capabilities, CI/CD workflows, and user personalization features. These enhancements will advance NewsNuggets' goal of delivering intelligent, accessible, and user-driven news experiences powered by AI.

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